

MedPulse

Project Summary Report

Clinical Health Survey & Epidemiological Surveillance Platform

SAL Institute of Medical Sciences & Hospital, Ahmedabad

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1. Project Overview

What is MedPulse?

MedPulse is a Clinical Health Survey & Epidemiological Surveillance Platform for Medical Colleges built for SAL Institute of Medical Sciences & Hospital, Ahmedabad, Gujarat. It is a full-stack Node.js/Express/SQLite web application with 4 distinct user portals.

Technology Stack

- Backend: Node.js + Express.js (v5.2)
- Database: SQLite (better-sqlite3)
- Frontend: Vanilla HTML/CSS/JS (no frameworks)
- Architecture: MVC with auto-loaded modular schemas and routes
- Auth: JWT-less localStorage session with route guards
- PWA Support: Service worker, manifest, installable

Modules

Database: 6 schema modules (00-core, 10-crm, 20-hms, 21-hospital-contact, 22-geofence, 30-academic)

API: 13 route modules

Portals: 4 modules (student, admin, patient, hospital)

2. Student Portal

Features & Pages

- Profile Page: Student identity card, SAL college info, NMC attendance data
- Schedule Page: Academic teaching schedule with horizontal cubic date cards
- Exams Page: Multi-year exam results, Paper 1/Paper 2
- Home Page: Community Medicine dashboard landing
- Family Manager: Family survey data, member health profiles
- Data Entry: Multi-step form for field health surveys
- Analytics: SQL-powered analytical reports and data queries
- Login/Register: Roll number + PIN authentication

3. Admin Portal

Features & Dashboard

- Dashboard: Faculty administrative overview
- Students Roster: Manage all medical students, batch filters
- Families & Surveys: View student-submitted survey data
- Attendance Sheet: Faculty-side datewise attendance marking
- Exams & Results: Enter/edit marks for all students
- Campaign CRM: Health campaign management
- Settings: Admin configuration

4. Patient Portal

Patient Features

- Overview: Patient health dashboard with vitals summary
- My Health Card: Digital patient identity and health card
- My Family: View family members from community survey data
- Call Hospital: Direct hospital contact integration
- Health Camps: Patient engagement campaigns
- Health Records: Complete medical records and visit history

5. Hospital Portal (FAP Hospital Control Centre)

Hospital Control Centre

- Control Centre: FAP overview with health indicator metrics
- FAP Patient Registry: Search & filter surveyed patients by condition
- Registered Patients: Hospital admission/OPD tracking with 4-step registration
- Colleges & Students: Student care team network view

6. Database & API Architecture

Architecture Highlights

- 6 Schema Modules: 00-core, 10-crm, 20-hms, 21-hospital-contact, 22-geofence, 30-academic
- 13 API Route Modules covering all application domains
- Geofencing integrated for field data collection
- Auth guard system supporting 4 distinct user types

MedPulse

Built for SAL Institute of Medical Sciences & Hospital

